# CBSE Class 10 Mathematics — Coordinate Geometry

**Time Allowed: 3 Hours** **Maximum Marks: 80**

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**General Instructions:**

1. This question paper contains 24 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

## Section A: Multiple Choice & Assertion-Reason (1 mark each)

**1**. The distance of the point $P(3, 4)$ from the origin is

(A) (A) 3
(B) (B) 5
(C) (C) 7
(D) (D) 25

**2**. The coordinates of the midpoint of the line segment joining the points $A(-2, 8)$ and $B(-6, -4)$ are

(A) (A) $(-4, -6)$
(B) (B) $(2, 6)$
(C) (C) $(-4, 2)$
(D) (D) $(-8, 4)$

**3**. The point on the $x$-axis which is equidistant from points $A(2, -5)$ and $B(-2, 9)$ is

(A) (A) $(-7, 0)$
(B) (B) $(0, -7)$
(C) (C) $(7, 0)$
(D) (D) $(0, 7)$

**4**. If the point $P(k, 0)$ divides the line segment joining $A(2, -2)$ and $B(-7, 4)$ in the ratio $1 : 2$, the value of $k$ is

(A) (A) 2
(B) (B) $-2$
(C) (C) 1
(D) (D) $-1$

**5**. The distance between the points $P(a, 0)$ and $Q(0, b)$ is

(A) (A) $a+b$

(B) (B) $\sqrt{a^2+b^2}$

(C) (C) $a^2+b^2$

(D) (D) $\sqrt{a+b}$

**6**. Assertion (A): The distance of the point $P(3,4)$ from the origin is 5 units. Reason (R): The distance of a point $(x,y)$ from the origin $(0,0)$ is given by $\sqrt{x^2+y^2}$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

**7**. Assertion (A): The midpoint of the line segment joining $A(2,4)$ and $B(4,2)$ is $(3,3)$. Reason (R): The midpoint of a line segment with endpoints $(x_1,y_1)$ and $(x_2,y_2)$ is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

**8**. Assertion (A): The point $(0,5)$ lies on the $y$-axis. Reason (R): For any point on the $y$-axis, the $x$-coordinate is zero.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

**9**. Assertion (A): The distance between the points $A(0,0)$ and $B(0,7)$ is 7 units. Reason (R): The distance between two points $(x_1,y_1)$ and $(x_2,y_2)$ is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

**10**. Assertion (A): The point $(3, 0)$ lies on the $x$-axis. Reason (R): For any point on the $x$-axis, the $y$-coordinate is zero.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

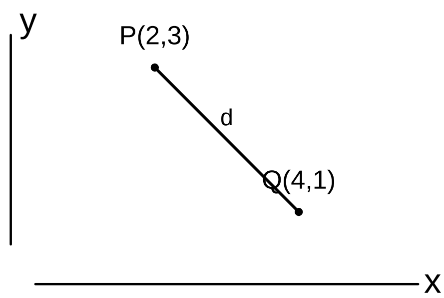
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

## Section C: Short Answer I (3 marks each)

**1**. Find the distance between the points $P(2, 3)$ and $Q(4, 1)$.

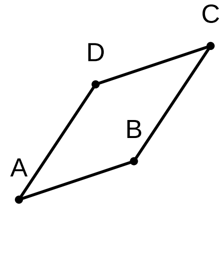



**2**. Determine the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$.

**3**. Find the coordinates of the midpoint of the line segment joining the points $A(-5, 4)$ and $B(7, -8)$.

**4**. Find the value of $k$ if the point $P(0, 2)$ is equidistant from $A(3, k)$ and $B(k, 5)$.

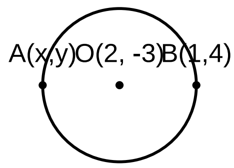
**5**. If the coordinates of the vertices of a parallelogram $ABCD$ are $A(1, 2)$, $B(4, 3)$, $C(6, 6)$, find the coordinates of the fourth vertex $D(x, y)$.



## Section D: Long Answer (4-5 marks each)

1. Find the ratio in which the $y$-axis divides the line segment joining the points $A(5,-6)$ and $B(-1,-4)$. Also, find the point of intersection.

2. Find the coordinates of a point $A$, where $AB$ is the diameter of a circle whose centre is $(2,-3)$ and $B$ is $(1,4)$.

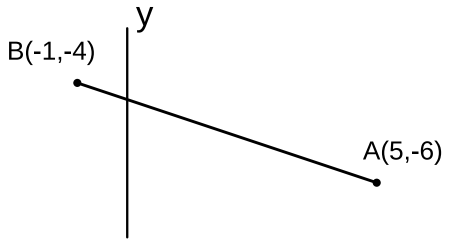



3. Find the value of $k$ for which the points $A(2,3)$, $B(4,k)$, and $C(6,-3)$ are collinear.

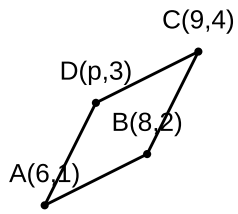
4. Find the distance between the points $P(-6,7)$ and $Q(-1,-5)$.

5. If the points $A(6,1)$, $B(8,2)$, $C(9,4)$, and $D(p,3)$ are the vertices of a parallelogram taken in order, find the value of $p$.

6. Find the ratio in which the $y$-axis divides the line segment joining the points $A(5,-6)$ and $B(-1,-4)$. Also, find the point of intersection.




7. If the points $A(6,1)$, $B(8,2)$, $C(9,4)$, and $D(p,3)$ are the vertices of a parallelogram taken in order, find the value of $p$.




**8**. Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$, and $(-3, -5)$.

**9**. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.